

31338 Network Servers

Lab 7c: Implementing backups with cron

Aims

1. Be able to use cron for scheduling regular tasks
2. To use standard UNIX tools to implement a simple backup process

Task 1: Using cron

Create a simple cron job that will create an empty file in the `/tmp` directory (using the `touch` command) at a predefined time. To understand the format of the information required in a `crontab` file, type: `man 5 crontab` (the "5" forces the command to be searched for in section 5 of the manual – section 5 contains descriptions of system file contents). Schedule your cron job to execute a few minutes later than the current time (at least 2 minutes later than the current time). Test that your cron job is working by looking for the empty file in the `/tmp` directory.

Note that there are two places you can put your cron job – either in `/etc/crontab` by using a normal text editor, or in the root user's own personal cron table. In the `/etc/crontab` file, you need to specify the username under which the job will be executed, but the username is not used in a user's personal cron table. The `crontab` command is used to edit/view/remove a user's personal cron table:

```
crontab -e      (edit)
crontab -l      (list)
crontab -r      (remove)
```

The editor used by default is `vi`. If you wish to use the `nano` editor (or `gedit` in a graphical environment), before you run any of the `crontab` commands, at your shell prompt, enter:

```
export EDITOR=nano
```

to set the value of the `EDITOR` environment variable equal to the name of the editor you wish to use by default.

Try creating cron jobs in both ways – using `/etc/crontab` and root's personal `crontab` file. Remember to test that they work.

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Task 2: Backup and restore

Your task is to backup two directories on your system, using two different backup tools. An introduction to the commands and their options is in the lecture slides – for more detail, refer to the man pages.

First, create a backup of the `/etc` directory using `tar`, and store the backup in a file named `/tmp/backup-etc.tar`

Secondly, create a backup of the `/opt` directory using `cpio`, and store the backup in a file named `/tmp/backup-opt.cpio`

View a table of contents of each of the archives to verify that the backups were created successfully.

Note that this is not the way backups are really implemented. Normally you backup an entire filesystem at once. We don't do that because we don't have enough spare disk space to store a copy of an entire filesystem backup. Also, normally your backup is stored on a tape device (or other media), so you would write the backup to `/dev/tape` or similar. We don't do that because we don't have a tape device or other removable media big enough (in theory we could do a small backup with a USB hard drive).

Now, from your `tar` backup, restore the file `/etc/hosts` into your (root's) home directory, just to test that you can do so.

It is important to always test that you can restore a file from your backups – you don't want to find out that your backup is corrupted when you really need to restore a file.

Task 3: Automating backups

Finally, combine what you have done in Tasks 1 and 2. Schedule your two backups (of `/etc` and `/opt`) to be performed once a week using cron. Check next week to verify that your backup is still occurring.